THE $K3 \times E$ CALABI–YAU THREEFOLD AND THE GENERATOR PENTAGON

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Abstract. For $X = K3 \times E$, the compact total-space traces, the Heisenberg specialisation, the automorphic BKM target, and the K3 fibre invariant separate: $\varkappa_{\mathrm{ch}}(X) = 0$, $\varkappa_{\mathrm{cat}}(X) = 0$, $\varkappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$, $\varkappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = 5$, and $\varkappa_{\mathrm{fiber}} = 24$. The six constructions attached to $K3 \times E$ have distinct mathematical data: R_1 is the Φ_3 specialisation under the stated CY_3 trace, framing, and specialisation hypotheses, R_2 is the automorphic Borcherds denominator target attached by the character-level arrow β_{23}, and $R_3, \ldots, R_6$ are independent chiral vertices. When the source-side bridges exist, the resulting diagram is a five-vertex generator pentagon with a distinguished automorphic source. The values $\{0, 3, 5, 24\}$ are subscripted invariant data. The numbers $\rho^{R_i} \in \{3, 12, 24\}$ below are generator ranks, never chiral invariant values. The Igusa normalisation is $D_5 = 64^{-1}\Delta_5$, with Δ_5 theta-normalised and D_5 monic; the imported BKM denominator is $D_5(2Z) = 64^{-1}\Delta_5(2Z)$. The primitive quotient-by-E OP/DT normalisation is $Z^X_{\mathrm{OP}} = Z^X_{\mathrm{DT}} = -4096\,\Delta_5^{-2}$; these scalar identities do not construct a compact Hall source.

1. The invariant spectrum

Proposition 1. *For $X = K3 \times E$,*

$$\varkappa_{\mathrm{ch}}(X) = 0, \qquad \varkappa_{\mathrm{cat}}(X) = \chi(\mathcal{O}_X) = 0.$$

The separate Heisenberg–Mukai, automorphic, and fibre constructions give

$$\varkappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3, \qquad \varkappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = 5, \qquad \varkappa_{\mathrm{fiber}} = 24.$$

Proof. Künneth gives $h^{0,\bullet}(K3 \times E) = (1, 1, 1, 1)$, so the alternating sum is 0. The same multiplicativity gives $\chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\chi(\mathcal{O}_E) = 2 \cdot 0 = 0$. The Heisenberg–Mukai specialisation uses a different trace readout: the $K3$ Mukai trace contributes 2, and the elliptic Heisenberg line contributes 1. Hence $\varkappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$. The fibre value is the topological Euler characteristic, equivalently the rank of the Mukai lattice $\widetilde{H}(K3, \mathbb{Z})$, equal to 24; it is not $\chi(\mathcal{O}_{K3}) = 2$. The Gritsenko–Nikulin denominator has $c_1(0) = 10$, so its Borcherds weight is $\varkappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = c_1(0)/2 = 5$. This is an automorphic target invariant, not a trace of $\mathrm{Perf}(X)$. □

2. Automorphic denominator

Definition 2 (Routes). Fix $X = S \times E$, with S a K3 surface. The six labels record different constructions, not six applications of Φ_3.

- R_1 the direct Φ_3 specialisation of $\mathrm{Perf}(X)$, conditional on the CY_3 smoothness, properness, negative-cyclic trace, and chain-level S^3-framing hypotheses, together with a witnessed (Σ_2, C) datum;
- R_2 the automorphic Borcherds denominator target, attached by the character-level arrow $\beta_{23} : A_X^{R_2,\mathrm{char}} \to A_X^{R_3}$, controlled by Δ_5, $D_5(2Z)$, and $c_1(0) = 10$;
- R_3 the Mukai lattice VOA construction;
- R_4 the Kummer/orbifold construction;
- R_5 the half-twisted sigma-model construction;

Date: 2026-04-30.

R_6 the BLLPR/Costello–Li Schur-sector construction.

The pentagon vertices are R_1, R_3, R_4, R_5, R_6; R_2 is the automorphic character source for target-side data, not a CY source, and does not by itself give a compact chiral vertex, a compact Hall source, or a ρ-rank entry.

Theorem 3 (Automorphic weight). *For the CHL/Gritsenko–Nikulin Borcherds-product levels $N \in \{1,2,3,4,6\}$, let $\mathcal{B}_N$ denote the automorphic Borcherds product and let $c_N(0)$ be the constant coefficient of the weight-zero weak Jacobi input. Then*

$$\kappa_{\mathrm{BKM}}(\mathcal{B}_N) = \frac{c_N(0)}{2}.$$

In particular $c_1(0) = 10$ gives $\kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = 5$ for the Gritsenko denominator. The symbol $\mathcal{B}_N$ is target-side automorphic notation; it is not an application of the CY-to-chiral functor to a compact Hall source.

Proof. Borcherds' automorphic-product theorem gives the weight $c_N(0)/2$; the Gritsenko–Nikulin specialisation fixes the genus-two variables, product cone, multiplier, and normalisation. At $N = 1$, the input is $\phi_{0,1}$ with $c_1(0) = 10$, the theta-normalised Gritsenko–Nikulin output is Δ_5, and the monic Borcherds product is

$$D_5 = 64^{-1}\Delta_5.$$

The denominator identity for the imported BKM Lie superalgebra is

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

The argument $2Z$ records the passage from the product character $\exp(-\pi i(\alpha, z))$ to the Weyl–Kac denominator character $\exp(-2\pi i(\alpha, z))$. The coefficient $c_1(0) = 10$ gives weight 5. The character-level denominator reads that weight as the invariant $\kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5})$. This is target-side Borcherds arithmetic: it fixes the denominator, weight, character, and normalisation. Its source-side Hall, CoHA, braided, and chiral lifts require separate compact route bridges; the denominator alone supplies no compact Hall correspondences, orientation line, coproduct, Hopf pairing, radical quotient, PBW/no-extra-relations theorem, or centre/Fock evaluation. Levels outside $N \in \{1,2,3,4,6\}$ require separate denominator data and are outside this theorem. Non-CHL Clery–Gritsenko diagonal-divisor rows belong to a separate catalogue with their own cover-group data. □

Proposition 4 (Igusa–OP scalar and the compact-Hall obstruction). *Let $X = K3 \times E$. In the Igusa cusp-form normalisation,*

$$D_5 = 64^{-1}\Delta_5,$$

where Δ_5 is theta-normalised and D_5 is the monic Borcherds product. The BKM denominator attached to $\mathfrak{g}_{\Delta_5}$ is

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

In the primitive quotient-by-E, Behrend-weighted OP/DT normalisation, the monic scalar product denoted χ_{10}^{OP} or Φ_{10}^{OP} is

$$\chi_{10}^{\mathrm{OP}} = \Phi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2,$$

and the reduced scalar normalisation is

$$Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

The obstruction to promoting these scalar data to a compact Hall source is the tuple

$$\mathfrak{o}_{K3\times E} = (\mathfrak{o}_{\mathrm{orient}}, \mathfrak{o}_{\mathrm{corr}}, \mathfrak{o}_{\mathrm{coprod}}, \mathfrak{o}_{\mathrm{pair}}, \mathfrak{o}_{\mathrm{rad}}, \mathfrak{o}_{\mathrm{PBW}}, \mathfrak{o}_{\mathrm{cent}}).$$

Its vanishing means, respectively: an orientation/Pfaffian line, compact Hall correspondences, a coproduct, a non-degenerate Hopf pairing, a closed radical quotient, PBW/no-extra-relations compatibility, and the *centre/Fock evaluation data have been constructed. The compact CoHA positive half is part of these source-side data: the correspondences must be proper on the completed support and compatible with the chosen orientation line. The displayed scalar identities do not force this vanishing.*

Proof. The equality $D_5 = 64^{-1}\Delta_5$ removes the theta-leading coefficient $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$. Squaring gives the OP monic scalar product $\chi_{10}^{\mathrm{OP}} = \Phi_{10}^{\mathrm{OP}} = D_5^2$, hence the normalisation $4096^{-1}\Delta_5^2$. The OP/DT scalar convention is $Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$, in the primitive quotient-by-E normalisation, which gives $-4096\,\Delta_5^{-2}$. These operations take place in the automorphic and scalar enumerative targets. A compact Hall source requires source-side objects: protected primitive representatives, compact extension correspondences, an orientation system, a coproduct, Hopf pairing matrices, radical descent, PBW/no-extra-relations compatibility, and centre/Fock evaluation. The automorphic form records the BKM denominator and the scalar character; it does not construct those source-side objects. In particular, 64, 4096, and the OP minus sign do not prove $\epsilon_o = \nu_{\Delta_5}$ for the compact orientation character. □

3. Generator pentagon

Definition 5 (Pentagon hypotheses)**.** The first four hypotheses are chain-level cells needed to strictify the $K3 \times E$ pentagon; composition is read right-to-left:

- P_1 the Costello–Li/BLLPR BV cell $\beta_{61} \circ \beta_{56} \simeq \beta_{51}$;
- P_2 the Kummer $\mathbb{Z}/2$-quotient cell $\beta_{45} \circ \beta_{34} \simeq \beta_{35}$;
- P_3 the half-twist cell $\beta_{56} \circ \beta_{45} \simeq \beta_{46}$;
- P_4 the HKR–Borcherds pentagonal filler $\beta_{61} \circ \beta_{56} \circ \beta_{45} \circ \beta_{34} \circ \beta_{13} \simeq \mathrm{id}_{A_X^{R_1}}$.

Here $\beta_{ij} : A_X^{R_i} \to A_X^{R_j}$, and $\beta_{51}, \beta_{35}, \beta_{46}$ denote the displayed composites. The two source-side assembly hypotheses are:

- P_5 finite-colimit representability of the five vertices in the completed presentable E_1-chiral category;
- P_6 compatibility of compact Hall/CoHA positive halves, proper Hall correspondences, orientation lines, Hopf pairings, radical quotients, Drinfeld doubles, and derived centres under the five bridges.

The automorphic data of R_2, including $D_5(2Z)$ and the OP/DT normalisation, are target-side inputs to these hypotheses, not proofs of them.

Theorem 6 (Pentagon ranks under bridge hypotheses)**.** *Assume the CY_3 trace/framing data for R_1, the five chiral vertices R_1, R_3, R_4, R_5, R_6, and the bridge hypotheses* $\mathrm{P}_1, \dots, \mathrm{P}_6$. *Then the generator ranks are*

$$\rho^{R_1} = 3, \quad \rho^{R_3} = 24, \quad \rho^{R_4} = 12, \quad \rho^{R_5} = 3, \quad \rho^{R_6} = 3.$$

Here $\rho^{R_i} := \mathrm{rank}_{\mathbb{Z}}\,\mathrm{Gen}(A_X^{R_i})$ is the rank of the chosen generator lattice in the i-th constructed chiral vertex. The bridge diagram has vertices R_1, R_3, R_4, R_5, R_6. The route R_2 supplies the automorphic BKM character source for β_{23}; the pentagon counts the five chiral vertices and assigns no ρ-rank to R_2. This is a rank statement, not a compact realisation theorem for $G(K3 \times E)$.

Proof. The route R_1 is the direct Φ_3 specialisation under the CY_3 smoothness, properness, negative-cyclic trace, chain-level S^3-framing, and witnessed-specialisation data. The Mukai-lattice vertex construction gives R_3, the Kummer/orbifold construction gives R_4, the half-twisted sigma model gives R_5, and the BLLPR/Costello–Li Schur-sector construction gives R_6. Each arrow in the pentagon is a bridge theorem: it must identify the stated generator lattice, preserve the E_1-chiral operation, and transport the relevant trace. The automorphic denominator in R_2 controls the Borcherds weight through $c_1(0)/2$ and the denominator identity $D_5(2Z) = 64^{-1}\Delta_5(2Z)$. It is not a sixth compact chiral vertex and it does not supply the compact Hall structure; those data are precisely the vanishing of $\mathfrak{o}_{K3\times E}$. At generator-rank level the arrows, in the order $\beta_{13}, \beta_{34}, \beta_{45}, \beta_{56}, \beta_{61}$, have types

$$3 \hookrightarrow 24, \qquad 24 \twoheadrightarrow 12, \qquad 12 \twoheadrightarrow 3, \qquad 3 \xrightarrow{\sim} 3, \qquad 3 \xrightarrow{\sim} 3.$$

Chain-level commutativity is exactly the content of $\mathrm{P}_1, \ldots, \mathrm{P}_6$, not a consequence of the scalar automorphic normalisation. Identification with a compact $G(K3 \times E)$ is a further compact-realisation theorem and requires $\mathfrak{o}_{K3\times E} = 0$. □

Conjecture 7. *Let* Pent_5 *be a five-cell Segal pentagon diagram in the completed presentable category* $\mathrm{ChirAlg}^{E_1}_{X,\infty}$. *Its vertices are* R_1, R_3, R_4, R_5, R_6, *and its arrows are*

$$\beta_{13},\ \beta_{34},\ \beta_{45},\ \beta_{56},\ \beta_{61}.$$

The arrows are required to preserve the positive-half structures, compact Hall correspondences, orientation lines, Hopf pairings, Drinfeld doubles, centres, traces, and finite-colimit closure wherever those objects have been built. The $\beta_{23} : A_X^{R_2,\mathrm{char}} \to A_X^{R_3}$ *attachment is only a character-level attachment, not a compact Hall morphism. The* $(\infty, 1)$*-categorical colimit of* Pent_5 *is only the categorical pentagon object. Its chain-level identification with* $G(K3 \times E)$ *is conditional on the* CY_3 *smoothness, properness, negative-cyclic trace, chain-level* S^3*-framing, and witnessed-specialisation data; on* $\mathrm{P}_1, \ldots, \mathrm{P}_4$*; on* P_5 *finite-colimit representability for the five vertices; on* P_6 *compact Hall/CoHA compatibility, double assembly, and centre continuity; on the bridge maps above; and on* $\mathfrak{o}_{K3\times E} = 0$.